

SIMATS ENGINEERING ASSESSMENT TOOL 1 – ENUMERATION ANALYSIS

COURSE CODE: ITA14
COURSE NAME: ETHICAL HACKING

CO3: Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities.

TECHNICAL ENUMERATION ANALYSIS REPORT **Nmap-Based Network Service Enumeration and Security Analysis**

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1. Question 1 – scanme.nmap.org

The assessment requires host discovery, TCP port scanning, service-version detection, OS fingerprinting and default script scanning, followed by business-purpose, attack-surface, threat and risk analysis.

Aim

To perform authorized Nmap enumeration of scanme.nmap.org, identify exposed services, analyze the attack surface, assess risk, and recommend appropriate security controls.

Nmap Commands

```
nmap -sn scanme.nmap.org
```

```
nmap -sT scanme.nmap.org
```

```
nmap -sV scanme.nmap.org
```

```
sudo nmap -O scanme.nmap.org
```

```
nmap -sC scanme.nmap.org
```

```
Firefox ESR
Scanme.nmap.org
kali@kali:~$ nmap -sn scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:04 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0030s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01:f83c:91ff:fe18:b02f
Nmap done: 1 IP address (1 host up) scanned in 4.78 seconds

kali@kali:~$ nmap -sT scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:05 -0400
failed to resolve 'scanme.nmap.org'.
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 2.55 seconds

kali@kali:~$ nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:05 -0400

kali@kali:~$ nmap -O scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:06 -0400
Command 'nmap' not found, did you mean:
command 'nmap' from deb nmap
command 'nmap' from deb nmap
command 'nmap' from deb nmap
command 'nmap' from deb nmap
command 'nmap' from deb python3-autobahn
Try: sudo apt install <deb name>

kali@kali:~$ nmap -sC scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:06 -0400
Status: 0:00:01 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host: Timing: About 0.00% done
Status: 0:00:02 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host: Timing: About 0.00% done
Status: 0:00:04 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host: Timing: About 0.00% done
Status: 0:00:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 9.82% done; ETC: 00:08 (0:01:32 remaining)

kali@kali:~$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:07 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0030s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01:f83c:91ff:fe18:b02f
```

```
Session Actions Edit View Help
kali@kali:~$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:07 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0030s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01:f83c:91ff:fe18:b02f
Not shown: 628 filtered tcp ports (no-response), 168 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9020/tcp   open  nmap-echo
31337/tcp  open  Elite
Device type: VoIP adapter (bridge) general purpose
Running (Just detected): AT&T embedded (88%), Oracle VirtualBox (88%), Sllrp (88%), QEMU (85%)
OS CPE: cpe:/o:oracle:virtualbox:cpel/danny.gaspardovskilrp:cpel/qemu:qemu
Aggressive OS guesses: AT&T embedded voice gateway (89%), Oracle VirtualBox Sllrp NAT bridge (88%), QEMU user mode network gateway (85%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 06.48 seconds

kali@kali:~$ nmap -sC scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:10 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0030s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01:f83c:91ff:fe18:b02f
Not shown: 613 filtered tcp ports (no-response), 82 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9020/tcp   open  nmap-echo
31337/tcp  open  Elite
ssh-hostkey:
|_ 2048 40:10:01:02:1f:cc:55:9d:0c:07:2b:36:97:0d:75:50a4 (RSA)
|_ 2048 2b:3d:2d:44:82:2a:b0:5a:9d:5b:03:05:14:c2:06:b2 (RSA)
|_ 256 00:00:02:b0:37:0a:1c:4b:83:2f:06:1c:1e:1a:02:37 (ECDSA)
|_ 256 33:fa:91:0f:0e:01:7b:1f:6d:05:a2:b0:f1:04:1a:56 (ED25519)
nmap-echo: open http
|_ http-title: Go Ahead and ScanMe!
|_ http-favicon: Nmap Project
9020/tcp   open  nmap-echo
31337/tcp  open  Elite
Nmap done: 1 IP address (1 host up) scanned in 185.77 seconds

kali@kali:~$ sudo nmap -O -sC -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:12 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.010s latency).
```

```
kali@kali: ~  
Session Actions Edit View Help  
31337/tcp open Elite  
Nmap done: 1 IP address (1 host up) scanned in 185.77 seconds  
[kali@kali]~  
$ sudo nmap -iS -O -v -sC scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:12 -0400  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.018s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 981 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)  
|_ ssh-hostkey:  
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)  
| 2048 28:3d:2d:a4:b2:22:10:81:5a:9d:53:b3:05:14:c2:e6:b2 (RSA)  
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)  
|_ 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)  
23/tcp    closed telnet  
25/tcp    closed smtp  
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))  
|_ http-title: Go ahead and ScanMe!  
|_ http-server-header: Apache/2.4.7 (Ubuntu)  
|_ http-favicon: Nmap Project  
111/tcp   closed rpcbind  
113/tcp   closed ident  
135/tcp   closed msrpc  
143/tcp   closed imap  
256/tcp   closed fad-secureremote  
995/tcp   closed pop3s  
1010/tcp  closed surf  
1720/tcp  closed h323qp31  
1900/tcp  closed mjpeg  
5900/tcp  closed vnc  
8888/tcp  closed sun-answerbook  
9029/tcp  open  nping-echo   Nping echo  
16012/tcp closed unknown  
31337/tcp open  tcpwrapped  
65532/tcp closed unknown  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (98%), Oracle Virtualbox (88%), Slirp (88%), QEMU (88%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/o:at&att_gosporovskii:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGM210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode network gateway (88%)  
No exact OS matches for host (test conditions non-ideal).  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 72.50 seconds  
[kali@kali]~
```

Analysis

Open ports identify externally reachable network entry points. Service names and versions provide information that an authorized security tester can use to prioritize further validation. Authentication-related services may attract credential attacks, while identifiable versions can be checked against vendor advisories during a controlled assessment.

Recommendations

- Disable unnecessary public services.
- Patch and securely configure required services.
- Restrict administrative services to trusted networks.
- Use strong authentication and monitoring.
- Apply firewall and network-access controls.

2. Question 2 – Metasploitable 2

The assessment identifies this address as Metasploitable 2 in an isolated laboratory environment and asks for comprehensive enumeration, service analysis, risk categorization, and discussion of potential entry points.

Aim

To perform authorized Nmap enumeration of 192.168.56.101, identify exposed services, analyze the attack surface, assess risk, and recommend appropriate security controls.

Nmap Commands

```
nmap -sn 192.168.56.101  
  
nmap 192.168.56.101  
  
nmap -p- 192.168.56.101  
  
nmap -sV 192.168.56.101  
  
sudo nmap -O 192.168.56.101
```

```
nmap -sC 192.168.56.101
```

```
sudo nmap -p- -sV -O -sC 192.168.56.101
```

```
kali@kali:~$ nmap -sC 192.168.56.101
Session Actions Edit View Help
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 72.50 seconds

kali@kali:~$ nmap -sC 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:16 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00039s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.52 seconds

kali@kali:~$ nmap 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:16 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00091s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 6.63 seconds

kali@kali:~$ nmap -p- 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:17 -0400
Stats: 0:00:28 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 16.52% done; ETC: 00:20 (0:02:06 remaining)

kali@kali:~$ nmap -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:19 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00066s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.84 seconds

kali@kali:~$ sudo nmap -O 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:20 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0016s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
```

```
kali@kali:~$ sudo nmap -O 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:20 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0016s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS811U print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP108 webcam (92%), Vantage HD710 OS satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.91 seconds

kali@kali:~$ nmap -sC 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:20 -0400
Stats: 0:00:04 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 32.80% done; ETC: 00:20 (0:00:04 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.0022s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 11.78 seconds

kali@kali:~$ sudo nmap -p- -sV -O -sC 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:21 -0400
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 6.64% done; ETC: 00:24 (0:02:49 remaining)
```

Analysis

The discovered services should be grouped by exposure and functionality. High-risk services may include vulnerable or unnecessarily exposed services; medium-risk services may disclose useful information or require additional conditions; low-risk services are generally necessary and appropriately restricted. Enumeration guides a tester in prioritizing controlled vulnerability validation.

Recommendations

- Remove unnecessary services.
- Update vulnerable software where applicable.
- Disable insecure legacy protocols.

- Restrict management interfaces.
- Use host firewall controls and network segmentation.

3. Question 3 – Windows Workstation

The assessment requires identification of active ports, network services, file-sharing protocols, remote-management services and operating-system details, followed by a least-privilege and hardening analysis.

Aim

To perform authorized Nmap enumeration of 192.168.56.102, identify exposed services, analyze the attack surface, assess risk, and recommend appropriate security controls.

Nmap Commands

```
nmap -sn 192.168.56.102
```

```
nmap -p- 192.168.56.102
```

```
nmap -sV 192.168.56.102
```

```
sudo nmap -O 192.168.56.102
```

```
nmap -sC 192.168.56.102
```

```
nmap -sV --script smb-os-discovery,smb-protocols 192.168.56.102
```

```

kali@kali:~$ nmap -sn 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:23 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00044s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.54 seconds

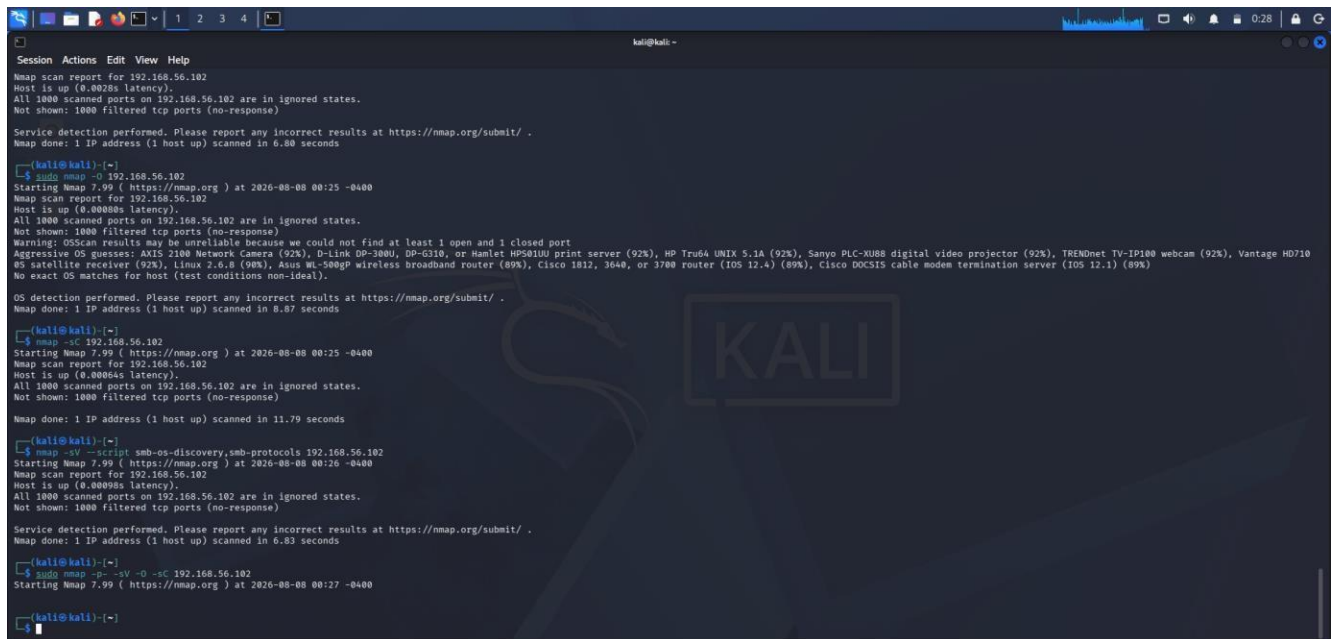
kali@kali:~$ nmap 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:23 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 6.63 seconds

kali@kali:~$ nmap -p- 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:24 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00051s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 6.67 seconds

kali@kali:~$ nmap -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:24 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0028s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.80 seconds

kali@kali:~$ sudo nmap -O 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:25 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00000s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: ASUS 2100 Network Camera (92%), D-Link DP-300U, DP-6310, or Hamlet HP810U print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD710
OS satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .

```



Analysis

Least privilege requires only necessary services to be exposed. File-sharing, remote-management, web or other network services should be checked against the workstation's intended role. Excessive exposure may support reconnaissance, credential attacks or lateral-movement opportunities in an authorized security test.

Recommendations

- Disable unnecessary network services.
- Restrict file sharing to required hosts/users.
- Disable insecure legacy protocols.
- Restrict remote administration.
- Apply security updates and host firewall rules.
- Segment sensitive resources from ordinary workstations.

4. Question 4 – OWASP Juice Shop Server

The assessment requires host discovery, port scanning, service/version detection, OS fingerprinting and web-service identification, followed by analysis of web exposure and supporting services.

Aim

To perform authorized Nmap enumeration of 192.168.56.103, identify exposed services, analyze the attack surface, assess risk, and recommend appropriate security controls.

Nmap Commands

```
nmap -sn 192.168.56.103

nmap -p- 192.168.56.103

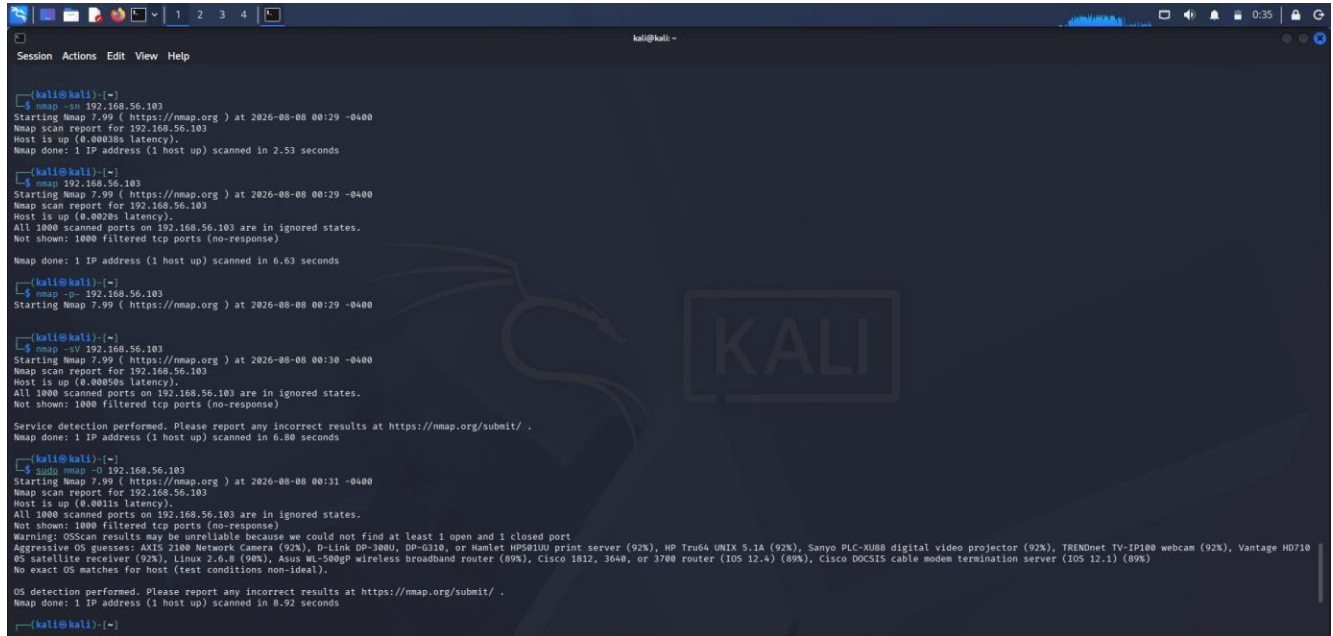
nmap -sV 192.168.56.103

sudo nmap -O 192.168.56.103

nmap -sC 192.168.56.103
```



```
nmap -sV --script http-title,http-headers 192.168.56.103
```



```
(kali@kali)~$ nmap -sn 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:29 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00038s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.53 seconds

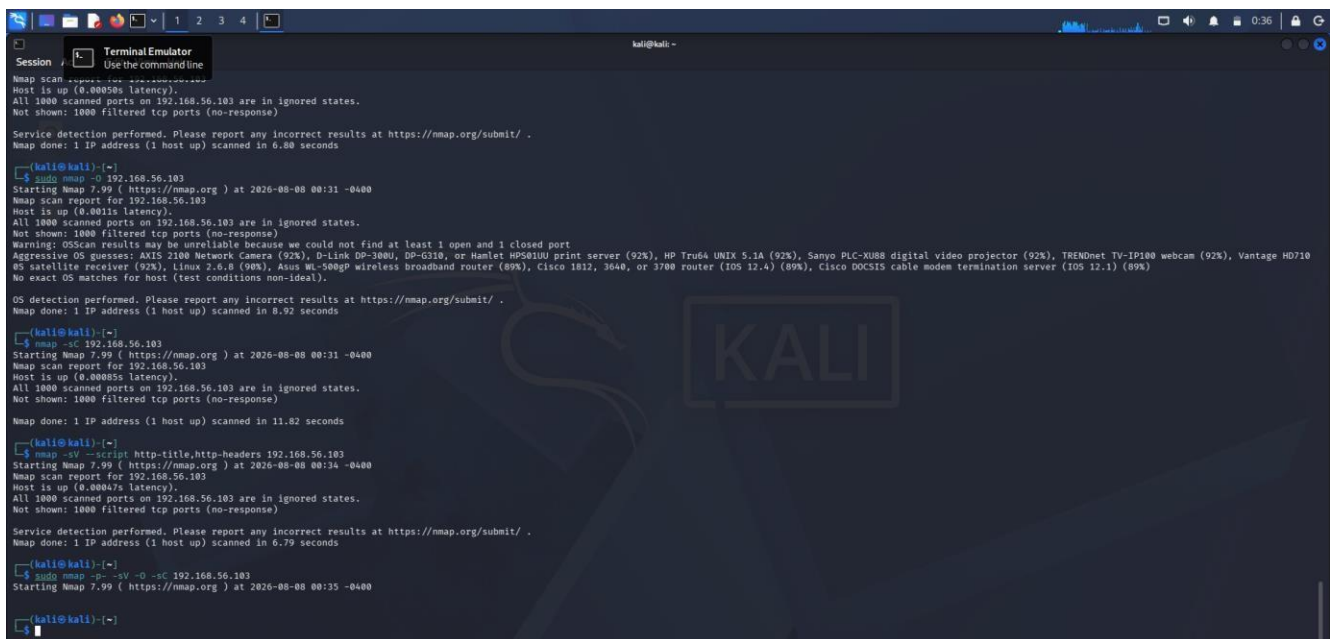
(kali@kali)~$ nmap 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:29 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0020s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 6.63 seconds

(kali@kali)~$ nmap -p- 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:29 -0400

(kali@kali)~$ nmap -sV 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:30 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00059s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.80 seconds

(kali@kali)~$ nmap -O 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:31 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD710
OS satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (105 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.92 seconds

(kali@kali)~$
```



```
(kali@kali)~$ nmap -sn 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:31 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00038s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.80 seconds

(kali@kali)~$ nmap -O 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:31 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-380U, DP-G310, or Hamlet HP5010U print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD710
OS satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (105 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.92 seconds

(kali@kali)~$ nmap -sC 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:31 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00055s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 11.82 seconds

(kali@kali)~$ nmap -sV --script http-title,http-headers 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:34 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00047s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.79 seconds

(kali@kali)~$ nmap -p- -sV -O -sC 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:35 -0400

(kali@kali)~$
```

Analysis

Web-service enumeration can reveal application-facing ports, server software, response information and supporting services. This information helps an authorized tester identify areas for later vulnerability assessment. Administrative interfaces and unnecessary supporting services should receive particular attention because they can enlarge the application's attack surface.

Recommendations

- Expose only required web services.
- Restrict administrative interfaces.
- Keep the web server and supporting software updated.
- Use secure transport and authentication controls.

- Apply network segmentation and monitoring.

5. Question 5 – Network Gateway

The assessment requires comprehensive enumeration of the laboratory gateway, including exposed management interfaces and services such as SSH, Telnet, SNMP, HTTP and HTTPS.

Aim

To perform authorized Nmap enumeration of 192.168.56.1, identify exposed services, analyze the attack surface, assess risk, and recommend appropriate security controls.

Nmap Commands

```
nmap -sn 192.168.56.1

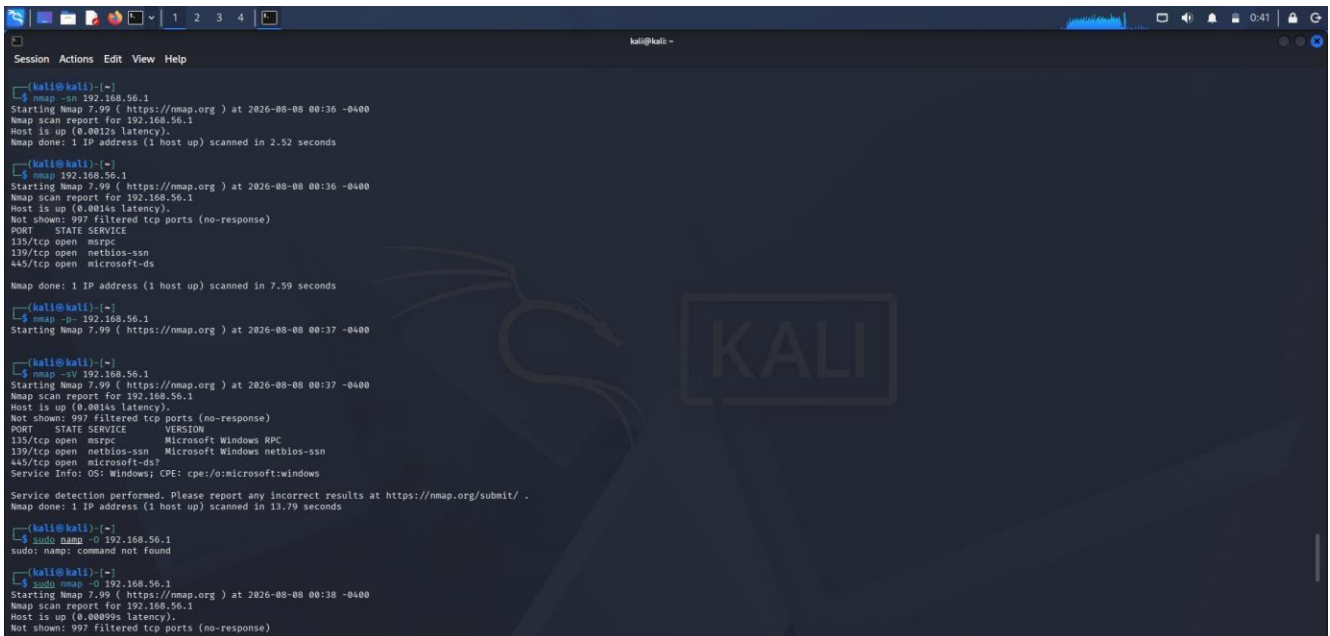
nmap -p- 192.168.56.1

nmap -sV 192.168.56.1

sudo nmap -O 192.168.56.1

nmap -sC 192.168.56.1

nmap -sV --script banner 192.168.56.1
```



```
kali@kali: ~
┌──(kali@kali)-[~]
│   └─$ nmap -sn 192.168.56.1
│       Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:36 -0400
│       Nmap scan report for 192.168.56.1
│       Host is up (0.0022s latency).
│       Nmap done: 1 IP address (1 host up) scanned in 2.52 seconds
│
│   └─$ nmap 192.168.56.1
│       Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:36 -0400
│       Nmap scan report for 192.168.56.1
│       Host is up (0.0014s latency).
│       Not shown: 997 filtered tcp ports (no-response)
│       PORT      STATE SERVICE
│       135/tcp    open  msrpc
│       139/tcp    open  netbios-ssn
│       445/tcp    open  microsoft-ds
│       Nmap done: 1 IP address (1 host up) scanned in 7.59 seconds
│
│   └─$ nmap -p- 192.168.56.1
│       Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:37 -0400
│
│   └─$ nmap -sV 192.168.56.1
│       Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:37 -0400
│       Nmap scan report for 192.168.56.1
│       Host is up (0.0014s latency).
│       Not shown: 997 filtered tcp ports (no-response)
│       PORT      STATE SERVICE      VERSION
│       135/tcp    open  msrpc        Microsoft Windows RPC
│       139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
│       445/tcp    open  microsoft-ds?
│       Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
│
│       Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
│       Nmap done: 1 IP address (1 host up) scanned in 13.79 seconds
│
│   └─$ sudo nmap -O 192.168.56.1
│       sudo: nmap: command not found
│
│   └─$ nmap -sV --script banner 192.168.56.1
│       Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:38 -0400
│       Nmap scan report for 192.168.56.1
│       Host is up (0.00099s latency).
│       Not shown: 997 filtered tcp ports (no-response)
```



```
Running (cpe:/o:oracle:virtualbox:92% (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovskii:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BOW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.49 seconds

(kali@kali) ~$ nmap -sC 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:38 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0013s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  marpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds

Host script results:
|_ smb2-time:
|   date: 2026-08-08T04:39:06
|_ start-date: N/A
|_ smb2-security-mode:
|   3.1.1:
|     Message signing enabled but not required
Nmap done: 1 IP address (1 host up) scanned in 21.94 seconds

(kali@kali) ~$ nmap -sV --script banner 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:39 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0015s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp   open  marpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 28.71 seconds

(kali@kali) ~$ sudo nmap -p- -sV -O -sC 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 00:40 -0400

(kali@kali) ~$
```

Analysis

Management interfaces can significantly increase the attack surface of a gateway. SSH should be restricted and strongly authenticated; Telnet should normally be disabled because it is a legacy insecure management protocol; SNMP should be restricted and securely configured; HTTP management should be avoided where secure HTTPS is available; and HTTPS administration should be protected with strong authentication and secure configuration.

Recommendations

- Disable Telnet when not required.
- Restrict SSH to trusted management systems.
- Secure and restrict SNMP.
- Prefer HTTPS for web administration.
- Apply ACLs/firewall restrictions.
- Update gateway firmware.
- Monitor administrative access.

9. Overall Risk Assessment

Risk ratings must be finalized using the actual Nmap observations. The following framework can be applied consistently across all five targets.

Risk	Typical Interpretation	Recommended Priority
High	Unnecessary or vulnerable exposed service, sensitive management interface, or significant attack-surface concern	Immediate remediation or strict isolation
Medium	Useful information exposure or service that requires additional conditions for abuse	Prioritize hardening and monitoring
Low	Necessary service with limited exposure and appropriate controls	Maintain secure configuration and monitor

10. Consolidated Security Recommendations

- Maintain an approved inventory of network services and their business purpose.
- Disable services that are not required for the system's intended role.
- Restrict management services to dedicated trusted administration networks.
- Patch operating systems, application servers and network devices regularly.
- Use strong authentication and least-privilege access.
- Disable insecure legacy protocols where possible.
- Apply host and network firewall rules.
- Use network segmentation to reduce lateral movement opportunities.
- Monitor logs and authentication events for suspicious activity.
- Perform periodic authorized enumeration and vulnerability assessments.

11. Conclusion

Nmap enumeration provides a practical method for identifying reachable hosts, open ports, network services, software versions and operating-system characteristics. Across the five assessment scenarios, the collected information forms the foundation for attack-surface analysis and risk-based security hardening.

The most important principle is that an exposed service should have a justified business or technical purpose and should be securely configured. Enumeration results should therefore be converted into actionable controls such as service reduction, patching, access restriction, segmentation, secure authentication and monitoring.

The final submission should include the student's actual Nmap screenshots and observed results in the evidence sections. This ensures that the report represents the real laboratory execution rather than assumed or fabricated scan output.